

IT1080C Computer Networking

Packet Tracer Introduction Lab

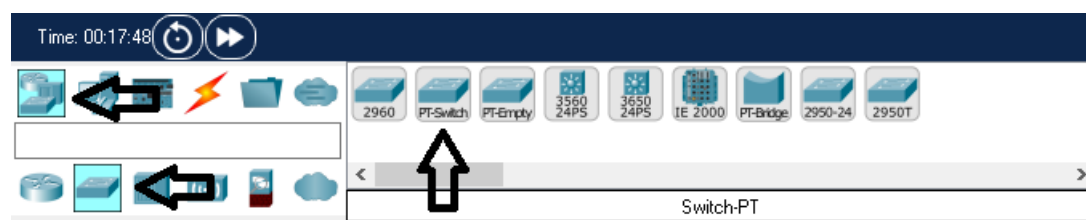
Creating a small network

Note: In Packet Tracer, the white screen is the workspace and icons are located on the bottom left. This lab uses a Switch and a Router from Network Devices; a Server and PC from End Devices; and straight-through cabling from Connections.

Add to workspace a PT-Router



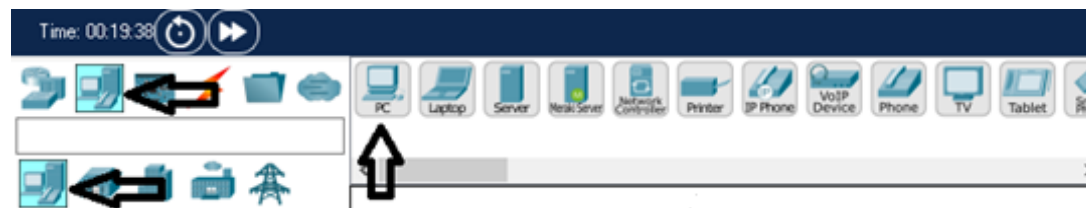
PT-Switch



Server



PC



Connector (straight-through cable)



Instructions

Add the devices – a router, a switch, a pc, and a sever to the workspace. Connect each device to the switch using straight-through cables.

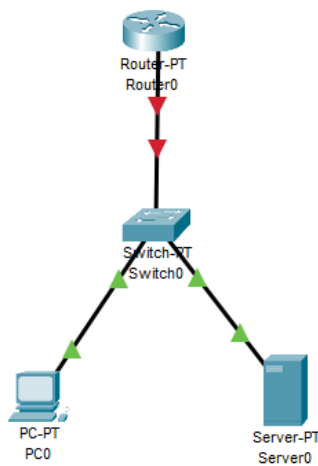


Figure 1 Connected Devices

The IP address range **192.168.0.0/24** is assigned to the local area network (LAN).

The range of addresses available is **192.168.0.0 – 192.168.0.255**. The subnet mask is **255.255.255.0**.

- **192.168.0.0** is the **network address** and cannot be assigned to a device.
- **192.168.0.255** is the **broadcast address** and cannot be assigned to a device.
- The first available address on the range is **192.168.0.1** which is typically **assigned to the default gateway**, the router link that connects to the LAN.
- The last available address on the range is **192.168.0.254** which is typically **assigned to the link to a server**.

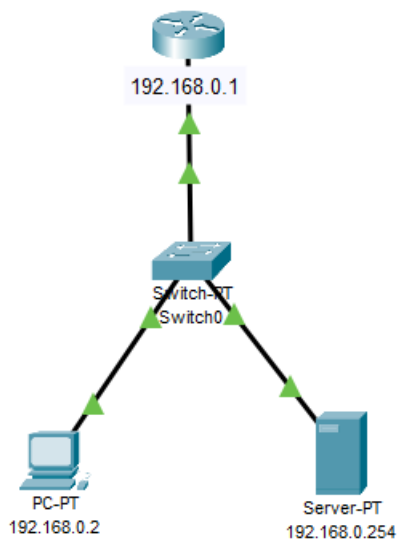


Figure 2 Diagram with IP Addresses

Configure the router with IP address 192.168.0.1 subnet mask 255.255.255.0.

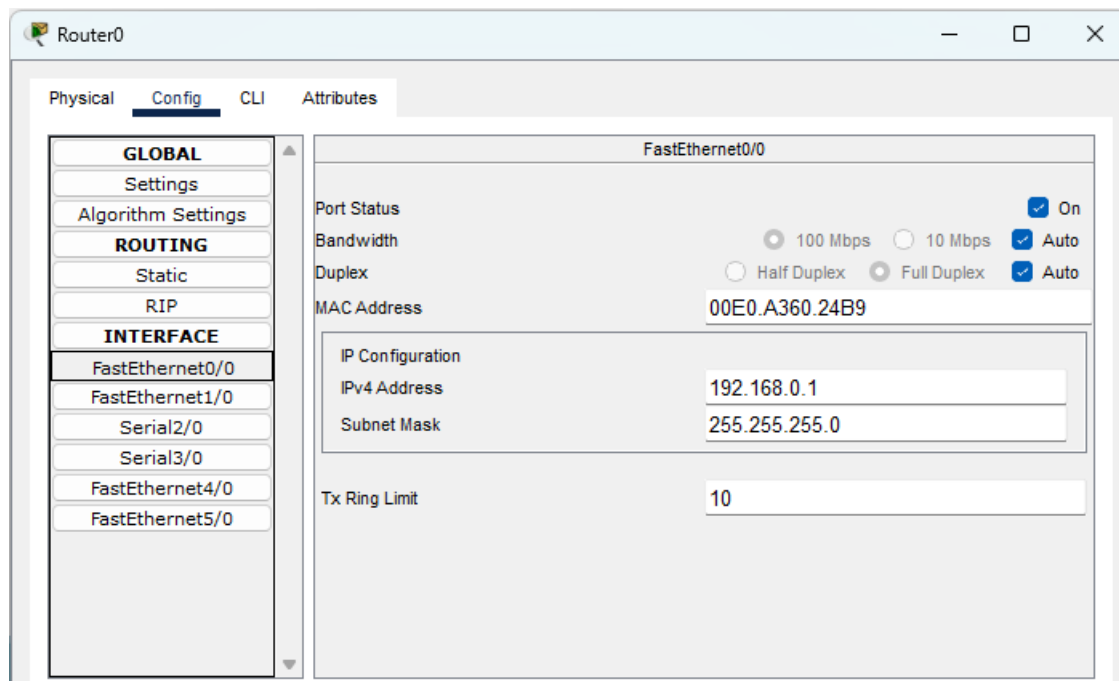


Figure 3 Router Configuration

Turn the port on and ensure duplexing is set to full. By default, router ports are shut down for security reasons.

Configure the server with IP address 192.168.0.254, subnet mask 255.255.255.0, and default gateway 192.168.0.1.

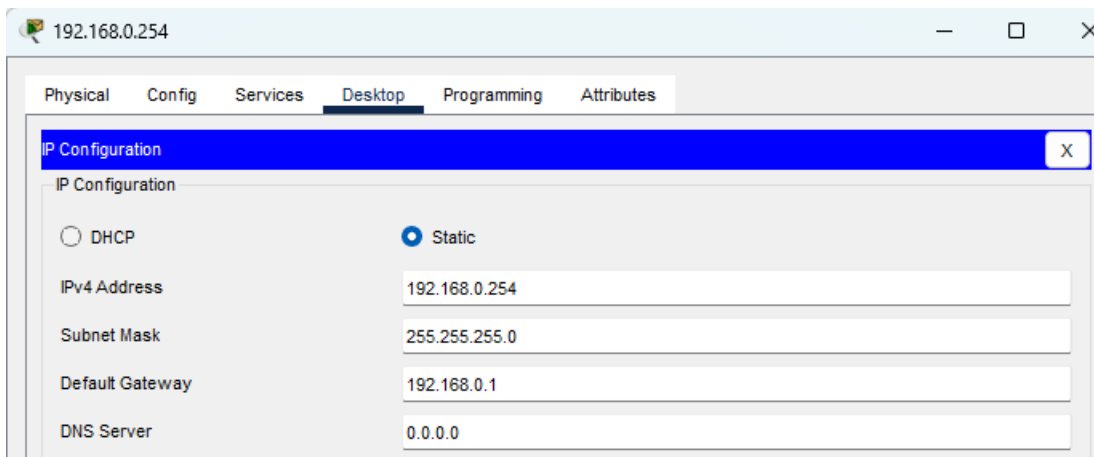


Figure 4 Server Configuration

Configure the pc with IP address 192.168.0.2, subnet mask 255.255.255.0, and default gateway 192.168.0.1.

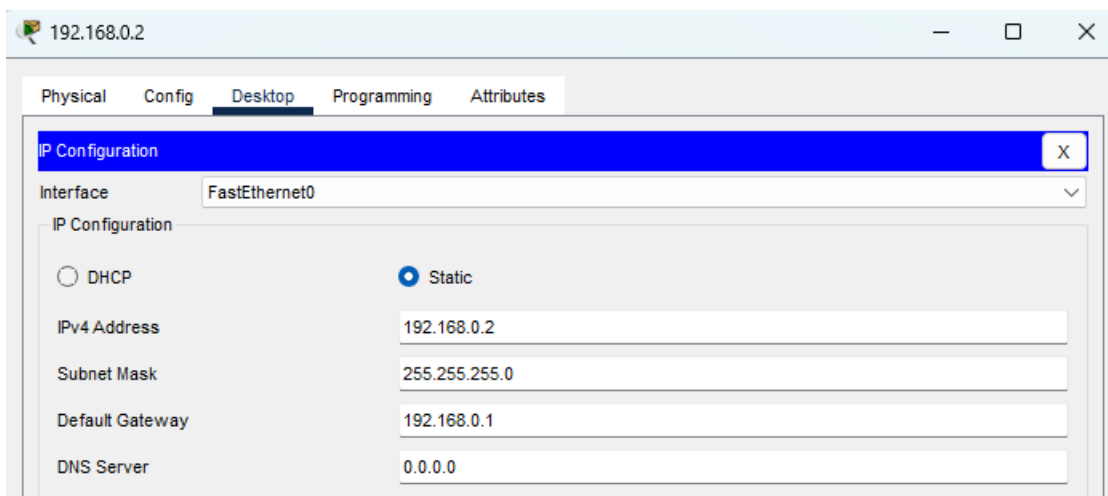


Figure 5 PC Configuration

Ping from the pc to the default gateway.

```
:\>ping 192.168.0.1
```

Pinging 192.168.0.1 with 32 bytes of data:

```
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
```

```
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping from the pc to the server.

```
C:\>ping 192.168.0.254
```

Pinging 192.168.0.254 with 32 bytes of data:

```
Reply from 192.168.0.254: bytes=32 time<1ms TTL=128
Reply from 192.168.0.254: bytes=32 time<1ms TTL=128
Reply from 192.168.0.254: bytes=32 time<1ms TTL=128
Reply from 192.168.0.254: bytes=32 time<1ms TTL=128
```

Ping statistics for 192.168.0.254:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

As seen on the command prompt of the pc.

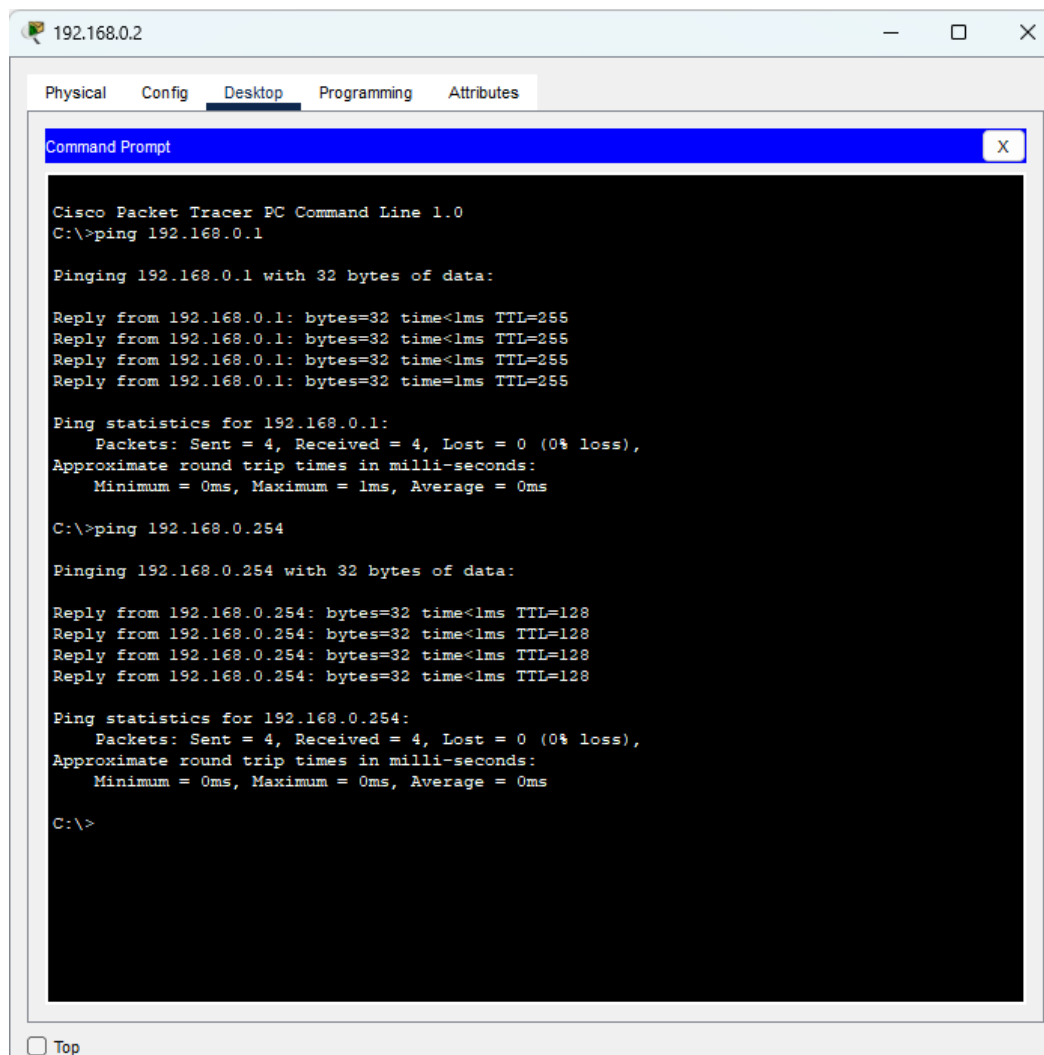


Figure 6 Command Prompt

Save the file using your name with the name of the lab, such as smithPTIntroLab.